

Logarithmic Curvature Flow, Filament Localization, and the Geometric Origin of the Lepton Mass Spectrum

Richard J. Reyes

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Abstract

The origin of the particle mass spectrum remains one of the central open problems in fundamental physics. Wave Confinement Theory models particles as localized curvature modes of a nonlinear complex wavefield. The curvature–feedback operator admits a logarithmic representation that transforms the governing dynamics into a viscous Hamilton–Jacobi equation. Through the Cole–Hopf transformation this nonlinear flow linearizes to a diffusion equation, implying that persistent localized structures cannot arise from local nonlinear trapping alone and instead require global topological confinement.

Curvature coupling in Frenet coordinates produces filamentary solutions whose core curvature invariant determines the rest mass scale of confined modes. Toroidal winding configurations generate a discrete curvature spectrum in which lower winding numbers produce larger curvature and therefore larger masses. If a closed confined filament supports a weakly modulated periodic curvature profile, then three sampled curvature modes embed on a circular manifold in $\sqrt{m}$ space, yielding the geometric structure of the Koide relation for charged leptons.

The logarithmic reduction identifies the universal local flow structure, while the existence of stationary modulated confined filaments depends on the full regularized WCT confinement problem rather than on the leading-order logarithmic flow alone.

1 Introduction

Wave Confinement Theory (WCT) proposes that particles correspond to stable localized curvature configurations of a nonlinear wavefield. In this framework the curvature of the wavefield determines an effective wave number that sets the rest energy scale of confined modes.

The goal of this work is to show that the curvature-feedback operator underlying WCT admits a logarithmic representation that naturally produces filament localization, curvature spectra, and a geometric interpretation of the charged lepton mass hierarchy.

2 Logarithmic Form of the Curvature Operator

Wave Confinement Theory introduces the nonlinear curvature operator

$$\Theta[\psi] = -\frac{\nabla^2\psi}{\psi + \varepsilon e^{-\alpha|\psi|^2}}. \quad (1)$$

For amplitudes sufficiently far from the regularization regime

$$|\psi| \gg \varepsilon e^{-\alpha|\psi|^2},$$

the denominator is dominated by ψ and the operator reduces to

$$\Theta[\psi] \approx -\frac{\nabla^2 \psi}{\psi}. \quad (2)$$

2.1 Logarithmic Field Transform

Introduce the logarithmic field

$$u = \ln \psi. \quad (3)$$

Then

$$\psi = e^u. \quad (4)$$

Taking derivatives gives

$$\nabla \psi = e^u \nabla u, \quad (5)$$

$$\nabla^2 \psi = e^u (\nabla^2 u + |\nabla u|^2). \quad (6)$$

Dividing by $\psi = e^u$ yields

$$\frac{\nabla^2 \psi}{\psi} = \nabla^2 u + |\nabla u|^2. \quad (7)$$

Thus the curvature operator becomes

$$\Theta[\psi] = -(\nabla^2 u + |\nabla u|^2). \quad (8)$$

2.2 Log-Curvature Evolution Equation

Assuming curvature-driven relaxation dynamics

$$\partial_t \psi = -\Theta[\psi] \psi, \quad (9)$$

we obtain

$$\partial_t \psi = \psi (\nabla^2 u + |\nabla u|^2). \quad (10)$$

Since $\psi = e^u$, we also have

$$\partial_t \psi = e^u \partial_t u. \quad (11)$$

Dividing by e^u yields

$$\boxed{\partial_t u = \nabla^2 u + |\nabla u|^2} \quad (12)$$

which is a viscous Hamilton–Jacobi equation.

2.3 Stationary Solutions

Stationary configurations satisfy

$$\nabla^2 u + |\nabla u|^2 = 0. \quad (13)$$

Using $\psi = e^u$, this condition becomes

$$\nabla^2 \psi = 0. \quad (14)$$

Thus equilibrium configurations correspond to harmonic wavefields except where topological constraints introduce singular structures.

3 Linearization via the Cole–Hopf Transform

The logarithmic curvature equation

$$\partial_t u = \nabla^2 u + |\nabla u|^2 \quad (15)$$

admits an exact linearization.

Define

$$\psi = e^u. \quad (16)$$

Then

$$\partial_t \psi = \frac{\partial_t \psi}{\psi}, \quad (17)$$

$$\nabla^2 u = \frac{\nabla^2 \psi}{\psi} - \frac{|\nabla \psi|^2}{\psi^2}. \quad (18)$$

Substituting yields

$$\partial_t \psi = \nabla^2 \psi. \quad (19)$$

Thus the nonlinear curvature flow is exactly equivalent to diffusion of the underlying wavefield.

3.1 Role of Topology in Confinement

On open domains the diffusion equation does not support stable localized structures, since diffusion spreads energy over time.

Localization therefore cannot arise from the local dynamics alone. Instead it must be enforced by global geometric or topological constraints on the domain of the wavefield.

Closed curves, periodic boundary conditions, or toroidal topology can restrict the diffusion process and allow stationary confined states to emerge. The filament solutions studied below should therefore be understood as curvature modes selected by the global topology of the wavefield domain.

4 Filament Reduction and Curvature Localization

Consider a localized solution around a closed curve $\Gamma(s)$ parameterized by arc length s .

The Laplacian in Frenet coordinates has the expansion

$$\nabla^2 = \partial_r^2 + \frac{1}{r} \partial_r + \frac{1}{r^2} \partial_\phi^2 + \partial_s^2 - \kappa \cos \phi \partial_r + O(r), \quad (20)$$

where κ is the curvature of the core curve. The curvature term $\kappa \cos \phi \partial_r$ couples neighboring angular modes $m \rightarrow m \pm 1$. For thin filaments we assume a dominant angular mode and neglect higher-order mode mixing.

Assume

$$u(r, \phi, s) = f(r) + im\phi + iks. \quad (21)$$

Then the stationary equation reduces to

$$f'' + \frac{1}{r} f' - \kappa f' + f'^2 = 0. \quad (22)$$

Localization occurs when

$$|\nabla u| \sim \kappa. \quad (23)$$

5 Curvature Scale and Energy

Define the curvature invariant of the filament core

$$\sigma = \sqrt{\kappa^2 + \tau^2}, \quad (24)$$

where κ is the curvature and τ the torsion of the central curve.

Identifying the curvature scale with the inverse Compton wavelength

$$\sigma = \frac{mc}{\hbar} \quad (25)$$

yields

$$m = \frac{\hbar}{c} \langle \sigma \rangle. \quad (26)$$

6 Toroidal Curvature Spectrum

For a helix of radius R and pitch parameter p , the curvature and torsion are

$$\kappa = \frac{R}{R^2 + p^2}, \quad \tau = \frac{p}{R^2 + p^2}. \quad (27)$$

The curvature invariant is therefore

$$\sigma = \sqrt{\kappa^2 + \tau^2} = \frac{1}{\sqrt{R^2 + p^2}}. \quad (28)$$

For thin tori ($r \ll R$) we identify the pitch with the winding parameter

$$p = n r, \quad (29)$$

where n is the winding number. Defining

$$\eta = \frac{r}{R}, \quad (30)$$

the curvature invariant becomes

$$\sigma_n = \frac{1}{R\sqrt{1 + n^2\eta^2}}. \quad (31)$$

Higher winding numbers increase the pitch, stretching the helix and reducing the curvature invariant. The $n = 0$ mode (a purely toroidal loop with no poloidal winding) carries the largest curvature $\sigma_0 = 1/R$.

Identifying σ_n with the inverse Compton wavelength gives the mass spectrum

$$m_n = \frac{\hbar \sigma_n}{c} = \frac{\hbar}{cR\sqrt{1 + n^2\eta^2}}. \quad (32)$$

The heaviest mode corresponds to the lowest winding number and the lightest mode to the highest winding number.

7 Variational Stability

The curvature energy functional is

$$E[\psi] = \int (|\nabla\psi|^2 + |\Theta[\psi]|^2) dx. \quad (33)$$

Stationary configurations correspond to extrema of this functional with respect to variations of the aspect ratio $\eta = r/R$.

8 Geometric Interpretation of the Koide Relation

The charged lepton masses satisfy

$$K = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3}. \quad (34)$$

Define

$$\mathbf{v} = (\sqrt{m_e}, \sqrt{m_\mu}, \sqrt{m_\tau}). \quad (35)$$

Koide showed that $\mathbf{v}$ lies on a circle in $\mathbb{R}^3$.

If three curvature eigenmodes generate the lepton masses, their square roots naturally form three points in $\sqrt{m}$ space that can be embedded on such a circle. Whether the toroidal curvature spectrum derived above produces this specific circular embedding remains an open problem requiring explicit evaluation of K as a function of η .

8.1 Periodic Curvature Spectrum and the Koide Geometry

8.2 Logarithmic Curvature Flow

Let the wavefield be

$$\psi(x, t). \quad (36)$$

Introduce the logarithmic transform

$$u = \ln \psi. \quad (37)$$

Substitution into the curvature operator yields the evolution equation

$$\partial_t u = \nabla^2 u + |\nabla u|^2, \quad (38)$$

which is the viscous Hamilton–Jacobi equation.

Applying the Cole–Hopf transformation

$$\psi = e^u \quad (39)$$

linearizes the dynamics to the diffusion equation

$$\partial_t \psi = \nabla^2 \psi. \quad (40)$$

Thus the nonlinear curvature dynamics are equivalent to diffusion in the underlying wave amplitude.

8.3 Closed Filament Reduction

Consider a localized state confined to a closed filament

$$\gamma(s) \quad (41)$$

parameterized by arclength s . For a loop of radius R , introduce the periodic coordinate

$$\phi = \frac{s}{R}, \quad \phi \in [0, 2\pi]. \quad (42)$$

Under the Cole–Hopf transformation, the leading-order logarithmic flow reduces to diffusion,

$$\partial_t \psi = \frac{1}{R^2} \frac{\partial^2 \psi}{\partial \phi^2}. \quad (43)$$

Pure diffusion on a periodic domain admits the Fourier solution

$$\psi(\phi, t) = \sum_{n=-\infty}^{\infty} a_n e^{-n^2 t / R^2} e^{in\phi}, \quad (44)$$

so that all nonzero modes decay as $t \rightarrow \infty$, leaving a spatially uniform state. Thus the leading-order Cole–Hopf reduction by itself does not sustain nontrivial stationary periodic structure.

Accordingly, any persistent nonuniform filament profile must arise from the full WCT confinement problem: namely, from the regularized curvature operator together with the global topological constraints of the closed filament. The role of the diffusion reduction is therefore to identify the local smoothing tendency, while the existence of a stationary modulated filament must be treated as a property of the full confined geometry rather than of the bare diffusion equation alone.

In what follows, assume that the closed filament supports a weakly modulated stationary envelope. The lowest harmonic compatible with periodicity is then

$$\psi(\phi) = A + B \cos \phi, \quad B \ll A, \quad (45)$$

which serves as the leading-order periodic ansatz for the confined mode.

8.4 Curvature Spectrum from the Periodic Ansatz

The leading curvature feedback scalar associated with the reduced operator is

$$\sigma(\phi) := -\frac{\partial_\phi^2 \psi}{\psi}. \quad (46)$$

For the ansatz

$$\psi(\phi) = A + B \cos \phi, \quad (47)$$

one has

$$\partial_\phi^2 \psi = -B \cos \phi, \quad (48)$$

hence

$$\sigma(\phi) = \frac{B \cos \phi}{A + B \cos \phi}. \quad (49)$$

For weak modulation $B \ll A$, expand

$$\frac{1}{A + B \cos \phi} = \frac{1}{A} \left(1 - \frac{B}{A} \cos \phi + O\left(\frac{B^2}{A^2}\right) \right), \quad (50)$$

which gives

$$\sigma(\phi) = \frac{B}{A} \cos \phi - \frac{B^2}{A^2} \cos^2 \phi + O\left(\frac{B^3}{A^3}\right). \quad (51)$$

Thus, to second order in the modulation parameter,

$$\sigma(\phi) \approx c_0 + c_1 \cos \phi + c_2 \cos^2 \phi. \quad (52)$$

Any quadratic trigonometric expression of this form can be written by completion of the square as

$$\sigma(\phi) \approx \sigma_* + (a + b \cos \phi)^2, \quad (53)$$

for suitable constants σ_*, a, b . Since only curvature differences are relevant to the mass-phase geometry, the constant offset σ_* may be absorbed into the background curvature scale, leaving the effective periodic curvature profile

$$\boxed{\sigma_{\text{eff}}(\phi) \approx (a + b \cos \phi)^2}. \quad (54)$$

8.5 Discrete Curvature Samples and Mass Geometry

Now sample the periodic curvature profile at three equally separated phase offsets

$$\phi_j = \delta + \frac{2\pi j}{3}, \quad j = 0, 1, 2. \quad (55)$$

This gives the three curvature values

$$\sigma_j = (a + b \cos \phi_j)^2. \quad (56)$$

Assuming the WCT mass law

$$m_j \propto \sigma_j, \quad (57)$$

one obtains

$$m_j = \mu(a + b \cos \phi_j)^2 \quad (58)$$

for some overall scale $\mu > 0$. Taking square roots,

$$\sqrt{m_j} = \sqrt{\mu} (a + b \cos \phi_j). \quad (59)$$

Therefore the square-root mass vector has the form

$$(\sqrt{m_0}, \sqrt{m_1}, \sqrt{m_2}) = \sqrt{\mu} \left(a + b \cos \delta, a + b \cos \left(\delta + \frac{2\pi}{3} \right), a + b \cos \left(\delta + \frac{4\pi}{3} \right) \right). \quad (60)$$

This is exactly the standard Koide-circle parameterization. Hence:

$$\boxed{\sqrt{m_j} = \sqrt{\mu} \left(1 + \sqrt{2} \cos \left(\delta + \frac{2\pi j}{3} \right) \right)} \quad (61)$$

for a suitable normalization of a and b , and consequently

$$\boxed{\frac{m_0 + m_1 + m_2}{(\sqrt{m_0} + \sqrt{m_1} + \sqrt{m_2})^2} = \frac{2}{3}}. \quad (62)$$

8.6 Conditional Closure

The derivation above proves the following conditional statement:

Proposition. *If a closed WCT filament supports a weakly modulated periodic stationary envelope whose leading-order profile is $\psi(\phi) = A + B \cos \phi$, and if the masses of the corresponding confined modes are proportional to the sampled effective curvature values of that filament, then the resulting mass triplet lies on the Koide circle and satisfies the charged-lepton Koide relation.*

Thus the remaining unresolved step is not the Koide geometry itself, but the full dynamical existence theory for the stationary modulated filament within the regularized WCT confinement operator.

9 Conditional Closure of the Koide-Saturated Lepton Family

We now isolate the part of the charged-lepton Koide geometry that can be closed within the present Wave Confinement Theory (WCT) framework.

The key point is that the WCT curvature harmonic fixes a distinguished mass-phase angle. Once the square-root mass vector is identified with the corresponding mass-phase vector, the full two-parameter family of Koide-saturated triplets follows. What remains undetermined is the overall mass scale and the phase on the Koide circle.

9.1 WCT Curvature Harmonic and the Koide Angle

For a stable spin- s toroidal eigenmode, WCT gives the curvature harmonic

$$K(s)^2 = \frac{2s}{2s+1}. \quad (63)$$

For a fermion mass triplet (m_1, m_2, m_3) , define the square-root mass vector

$$\mathbf{y} = \begin{pmatrix} \sqrt{m_1} \\ \sqrt{m_2} \\ \sqrt{m_3} \end{pmatrix}, \quad \mathbf{e}_0 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}. \quad (64)$$

Let θ be the angle between $\mathbf{y}$ and $\mathbf{e}_0$. Then Koide's geometric identity is

$$Q := \frac{m_1 + m_2 + m_3}{(\sqrt{m_1} + \sqrt{m_2} + \sqrt{m_3})^2} = \frac{1}{3 \cos^2 \theta}. \quad (65)$$

Assume now that the WCT curvature harmonic fixes the mass-phase angle via

$$\cos \theta = K(s). \quad (66)$$

Then

$$Q(s) = \frac{1}{3K(s)^2} = \frac{1}{3} \frac{2s+1}{2s} = \frac{2s+1}{6s}. \quad (67)$$

For charged leptons, $s = \frac{1}{2}$, so

$$K\left(\frac{1}{2}\right)^2 = \frac{1}{2}, \quad \cos \theta_\ell = \frac{1}{\sqrt{2}}, \quad \theta_\ell = \frac{\pi}{4}, \quad (68)$$

and hence

$$Q_\ell = \frac{2}{3}. \quad (69)$$

Thus, conditional on the identification $\cos \theta = K(s)$, WCT selects the charged-lepton Koide angle exactly.

9.2 General Form of All Koide-Saturated Triplets

We now determine the most general mass triplet satisfying $Q = \frac{2}{3}$.

Decompose the square-root mass vector into components parallel and orthogonal to $\mathbf{e}_0$:

$$\mathbf{y} = a \mathbf{e}_0 + b \mathbf{u}, \quad (70)$$

where

$$\mathbf{u} \cdot \mathbf{e}_0 = 0, \quad \|\mathbf{u}\| = 1. \quad (71)$$

Since the angle between $\mathbf{y}$ and $\mathbf{e}_0$ is $\theta_\ell = \pi/4$,

$$\cos \theta_\ell = \frac{a}{\sqrt{a^2 + b^2}} = \frac{1}{\sqrt{2}}, \quad (72)$$

which implies

$$a = b. \quad (73)$$

Therefore

$$\mathbf{y} = a(\mathbf{e}_0 + \mathbf{u}). \quad (74)$$

Choose an orthonormal basis of the plane orthogonal to $\mathbf{e}_0$:

$$\mathbf{e}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}. \quad (75)$$

Then every unit vector orthogonal to $\mathbf{e}_0$ can be written as

$$\mathbf{u} = \cos \delta \mathbf{e}_1 + \sin \delta \mathbf{e}_2, \quad (76)$$

for some phase angle δ .

Substituting this decomposition gives the standard Koide-circle parameterization

$$\sqrt{m_j} = \sqrt{\mu} \left(1 + \sqrt{2} \cos \left(\delta + \frac{2\pi j}{3} \right) \right), \quad j = 0, 1, 2, \quad (77)$$

for some scale $\mu > 0$. Squaring yields the full mass family

$$m_j = \mu \left(1 + \sqrt{2} \cos \left(\delta + \frac{2\pi j}{3} \right) \right)^2, \quad j = 0, 1, 2. \quad (78)$$

This is the complete set of all mass triplets saturating the charged-lepton Koide relation.

9.3 Interpretation within WCT

The present WCT derivation therefore closes to the following conditional statement:

$$\cos \theta_\ell = K\left(\frac{1}{2}\right) = \frac{1}{\sqrt{2}} \implies Q_\ell = \frac{2}{3} \implies m_j = \mu \left(1 + \sqrt{2} \cos \left(\delta + \frac{2\pi j}{3} \right) \right)^2. \quad (79)$$

The remaining undetermined quantities are:

- the overall scale μ , which sets the absolute mass scale,
- the phase δ , which selects the position of the triplet on the Koide circle.

Within WCT, these should be determined by the discrete shell-eigenvalue problem for the curvature operator on the charged-lepton topological sector. Equivalently, if the first three stable shell modes satisfy

$$\sqrt{\langle \sigma_{1,k_j} \rangle} = A + B \cos \left(\delta + \frac{2\pi j}{3} \right), \quad j = 0, 1, 2, \quad (80)$$

then, using the WCT mass law

$$m_j = \frac{\hbar}{c} \langle \sigma_{1,k_j} \rangle, \quad (81)$$

the charged-lepton masses obey the Koide relation exactly.

9.4 Theorem

Theorem. *Let a charged-lepton family correspond to a stable spin- $\frac{1}{2}$ WCT toroidal eigenmode with curvature harmonic $K^2 = \frac{1}{2}$. Assume further that the corresponding confined modes arise from sampled effective curvature values of a weakly modulated periodic filament profile. If the square-root mass vector is identified with the corresponding mass-phase vector, then the family necessarily satisfies the Koide relation*

$$Q_\ell = \frac{2}{3}, \quad (82)$$

and the most general such triplet is

$$m_j = \mu \left(1 + \sqrt{2} \cos \left(\delta + \frac{2\pi j}{3} \right) \right)^2, \quad j = 0, 1, 2. \quad (83)$$

Proof. The curvature-harmonic condition for $s = \frac{1}{2}$ gives

$$\cos \theta_\ell = K = \frac{1}{\sqrt{2}}, \quad (84)$$

hence

$$\theta_\ell = \frac{\pi}{4}. \quad (85)$$

Koide's geometric identity then gives

$$Q_\ell = \frac{1}{3 \cos^2 \theta_\ell} = \frac{2}{3}. \quad (86)$$

By the weakly modulated periodic filament hypothesis, the sampled effective curvature values have the cosine form

$$\sqrt{m_j} = \sqrt{\mu} \left(1 + \sqrt{2} \cos \left(\delta + \frac{2\pi j}{3} \right) \right), \quad j = 0, 1, 2, \quad (87)$$

which is precisely the general parameterization of all vectors in $\mathbb{R}^3$ making angle $\pi/4$ with the democratic axis. Squaring yields

$$m_j = \mu \left(1 + \sqrt{2} \cos \left(\delta + \frac{2\pi j}{3} \right) \right)^2, \quad j = 0, 1, 2. \quad (88)$$

Therefore the charged-lepton family satisfies the Koide relation and belongs to the two-parameter Koide-saturated mass family. $\square$

10 Correspondence with the KPZ Universality Class

The logarithmic curvature equation derived above

$$\partial_t u = \nabla^2 u + |\nabla u|^2 \quad (89)$$

belongs to a well-known class of nonlinear growth equations.

10.1 The KPZ Equation

The deterministic Kardar–Parisi–Zhang (KPZ) equation describing interface growth has the form

$$\partial_t h = \nu \nabla^2 h + \frac{\lambda}{2} |\nabla h|^2. \quad (90)$$

Here $h(x, t)$ represents the height of a growing interface, ν is a smoothing coefficient, and λ controls nonlinear slope amplification.

Identifying the logarithmic field u with the KPZ height field,

$$h \equiv u, \quad (91)$$

the WCT log-curvature evolution equation becomes

$$\partial_t u = \nabla^2 u + |\nabla u|^2 = \nu \nabla^2 u + \frac{\lambda}{2} |\nabla u|^2 \quad (92)$$

with the parameter identification

$$\nu = 1, \quad \lambda = 2. \quad (93)$$

Thus the leading-order WCT curvature relaxation flow lies in the *deterministic KPZ universality class*.

10.2 Cole–Hopf Linearization

A key property of the KPZ equation is the existence of the Cole–Hopf transformation. Defining

$$\psi = e^u, \quad (94)$$

the nonlinear equation

$$\partial_t u = \nabla^2 u + |\nabla u|^2 \quad (95)$$

maps exactly to the diffusion equation

$$\partial_t \psi = \nabla^2 \psi. \quad (96)$$

This identity demonstrates that the nonlinear curvature flow in logarithmic variables corresponds to linear diffusion of the underlying wavefield amplitude.

10.3 Interpretation within Wave Confinement Theory

The KPZ correspondence clarifies the role of curvature feedback in WCT. The two terms in the evolution equation

$$\partial_t u = \nabla^2 u + |\nabla u|^2 \quad (97)$$

have distinct geometric interpretations:

- $\nabla^2 u$ represents smoothing of the logarithmic amplitude, corresponding to diffusion of the underlying wavefield.
- $|\nabla u|^2$ represents nonlinear amplification of gradients, arising from curvature feedback between the Laplacian and the local field amplitude.

Together these terms produce curvature-driven growth dynamics belonging to the KPZ class.

10.4 Topological Confinement

On open domains the diffusion equation

$$\partial_t \psi = \nabla^2 \psi \quad (98)$$

does not support stable localized structures, since diffusion tends to spread energy over time.

Consequently, particle-like confinement in WCT cannot arise from local dynamics alone. Instead it must be enforced by global topological constraints such as closed loops, periodic boundary conditions, or toroidal geometry.

Within these geometries the KPZ-type curvature flow selects stable filamentary structures whose curvature scale determines the effective wave number and therefore the rest mass of the confined mode.

10.5 Regularized KPZ Structure

The full WCT curvature operator

$$\Theta[\psi] = -\frac{\nabla^2\psi}{\psi + \varepsilon e^{-\alpha|\psi|^2}} \quad (99)$$

introduces a nonlinear regularization that prevents singular behavior at small amplitudes. In the regime

$$|\psi| \gg \varepsilon e^{-\alpha|\psi|^2}, \quad (100)$$

the operator reduces to

$$\Theta[\psi] \approx -\frac{\nabla^2\psi}{\psi}, \quad (101)$$

recovering the exact KPZ/Cole–Hopf structure derived above. Thus the WCT curvature dynamics can be interpreted as a *regularized KPZ flow* governing the evolution of the logarithmic wavefield.

11 Conclusion

The curvature–feedback dynamics of the wavefield admit a logarithmic representation that converts the nonlinear evolution into a viscous Hamilton–Jacobi equation. Under the Cole–Hopf transformation this equation reduces to a diffusion equation for the underlying wave amplitude. As diffusion cannot sustain localized structures through local nonlinear trapping alone, persistent particle-like states must arise from global topological confinement of the wavefield.

Within such confined geometries the dynamics organize into filamentary curvature modes. The curvature invariant of the filament core sets the effective wave number of the confined state and therefore determines its rest mass scale. Toroidal winding configurations produce a discrete curvature spectrum in which smaller winding numbers correspond to larger curvature and therefore larger masses. If a closed confined filament supports a weakly modulated periodic curvature profile, then three sampled curvature modes embed on a circular manifold in $\sqrt{m}$ space, yielding the geometric structure of the Koide relation for the charged leptons.

The logarithmic reduction identifies the universal local flow structure, while the full existence theory for stationary modulated confined filaments remains a problem of the complete regularized WCT confinement operator.

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